

VAIBHAV SONI

Backend Developer

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Summary

Backend Developer with hands-on experience building RESTful APIs using Flask and FastAPI. Strong problem-solving mindset with a **generic** approach. Familiar with integrating Generative AI into backend systems for tasks such as data extraction, summarization, and content generation.

Skills

Programming: Python, SQL, JAVA

Backend Frameworks & APIs: FastAPI, Flask, RESTful API Development

Databases & Vector Stores: PostgreSQL, SQLite, ChromaDB, pgvector

AI Processing Tools & LLM Engineering:: LLMWhisperer (OCR), Whisper (Audio-to-Text), Roboflow (Vision tool), RAG, LangGraph, Prompt Engineering

Work Experience (1 Year)

Tata Consultancy Services

Jan 2025 – Current

Backend Developer

Kolkata, West Bengal

- Built multiple **GenAI** proof-of-concept applications using Streamlit to explore feasibility, user flow, and model performance before moving to production.
- Developed end-to-end GenAI solutions such as a Smart Claim Reader, Invoice Extractor (OCR + Roboflow), and an Insurance Agent Chatbot/ Call Agent/ Mail Agent, using pgvector with PostgreSQL for vector storage.
- Worked with LLaMA and Gemini as inference models and used MiniLM and Nomic as embedding models to support semantic search and retrieval-based workflows.

Education

Indian Institute of Information Technology, Madras

2023 – 2027

B.S (online degree) - Data Science and Application - CGPA - 7.97

Jodhpur, Rajasthan

Degree level (3rd year) - TEC (Total Earned Credits): 88

Mugneeram Bangur Memorial University

2020 – 2024

B.E - Electronics and Communication - CGPA - 8.01

Jodhpur, Rajasthan

Coordinator of King's Court Club (Official Chess Club)

Mahaveer Public School

2009 – 2019

Grade 12th - 83.2%

Jodhpur, Rajasthan

Grade 10th - 8.2 CGPA (77.9%)

PCM with Computer Science, House Captain in 12th Grade

Certifications

- Google Cloud Certified Generative AI Leader – Google Cloud (August 2025)
- Introduction to OpenAI API & ChatGPT API for Developers – Udemy (May 2025)
- Mastering Ollama: Build Private Local LLM Apps with Python – Udemy (May 2025)

Projects

BDM Project | *Data Science*

July 2025 – Nov 2025

- Designed and implemented a data-driven business analysis for a restaurant domain using 3 months of real data, covering sales-wise and item-wise performance metrics.
- Built structured master data and transactional datasets, performed data cleaning, aggregation, and analysis to generate business insights such as top-selling items, peak hours, and revenue trends.
- Created multiple analytical reports and visualizations (charts and summaries) to support management-level decision making, including a final presentation (PPT) and formal project documentation.

Engage 2 Value | *Python, Scikit-learn, EDA*

Jan 2025 – May 2025

- Built a machine learning model to predict customer purchase value using multi-session behavioral data across digital touchpoints, including device type, traffic source, browser, and geography..
- Performed end-to-end data preprocessing and feature engineering, handling large-scale datasets (116k+ rows) with a mix of categorical and numerical features, and engineered time-based features to improve signal quality.
- Trained and evaluated multiple models including Linear Regression, Random Forest, Gradient Boosting, and XGBoost, achieving the best performance with XGBoost (lowest RMSE) for final predictions and submission.

Quizgen | *Flask, Vue.js, Sqlite*

Jan 2025 – May 2025

- Developed a multi-user quiz management web application for exam preparation, supporting role-based access (Admin and Users) for managing courses, quizzes, and assessments.
- Built a RESTful backend using Flask and Flask-RESTful, integrated with SQLite via Flask-SQLAlchemy, and implemented secure authentication and authorization using Flask-Security.
- Improved performance and scalability by using Redis caching and Celery-based background tasks for **daily quiz reminders, monthly performance reports**, and exporting quiz results as CSV files.

Household Services Application | *Flask, Sqlite*

Sept 2025 – Dec 2024

- Developed a multi-user household services web application supporting Admin, Customer, and Service Professional roles for booking and managing home services.
- Built the backend using Flask and SQLAlchemy, designed relational database schemas with SQLite, and implemented service request workflows using date and time-based logic.
- Created role-specific dashboards with dynamic data visualization using Matplotlib, enabling admins and users to track service requests, ratings, and activity trends